

Poisoning the Threshold-Calibration Stage in Federated IoT Anomaly Detection: A Policy-Differentiated Vulnerability Analysis

Salaheddine Nasloubi¹, Mohamed Lachgar², and Hamid Hrimech¹

¹ LAMSAD Laboratory, National School of Applied Sciences (ENSA), Hassan First University, Berrechid, Morocco

² L2IS Laboratory, Higher Normal School (ENS), University Cadi Ayyad, Marrakech, Morocco

Abstract. Federated learning (FL) defenses for IoT anomaly detection focus on the training phase—Byzantine-robust aggregation, gradient filtering, differential privacy. The post-training *threshold-calibration stage*, where each client autonomously collects a benign calibration set and derives its detection threshold, is structurally outside the scope of these defenses. We characterize the vulnerability of this stage under a score-level contamination model: a single compromised client replaces a fraction f of its calibration buffer with tail-sampled benign reconstruction scores, shifting the detection threshold without altering model weights, training data, or test labels. Across three threshold-policy families (GLOBAL, LOCAL, CLUSTER) evaluated on the N-BaIoT nine-device federation, the score-level raising attack achieves 100% Gate-1 pass rate at all non-zero fractions, degrading victim true-positive rates by 2.4 to 7.9 percentage points across injection fractions (4.9–7.9 pp at $f=0.4$). The score-level lowering attack succeeds for LOCAL and CLUSTER (victim detection-rate increase of 10–12 pp as a lower bound, with coefficient-of-variation disparity $\Delta CV(FPR)$ increasing 0.05–0.06) but is structurally bounded for GLOBAL due to $1/N$ dilution. Policy choice determines not just per-victim magnitude but blast radius: LOCAL confines harm to the compromised client; GLOBAL dilutes per-victim shift but propagates it to all federation members.

Keywords: Federated Learning · Anomaly Detection · IoT Security · Poisoning Attack · Threshold Calibration · Vulnerability Analysis

1 Introduction

Federated learning (FL) has emerged as a natural fit for IoT anomaly detection: devices contribute to a shared model without exposing raw traffic, and personalized detection thresholds let each node adapt to its own traffic distribution [13,16]. The dominant security concern in federated anomaly detection focuses on the *training phase*—Byzantine clients injecting adversarial gradients, label-flipping poisoning, or model-inversion attacks [4,3,19]. Aggregation-phase

defenses such as FLTrust, Krum, and Trimmed-Mean target exactly this surface [6,5,20].

An undefended phase. What existing defenses collectively ignore is the *threshold-calibration stage*: the post-training, pre-deployment step in which each client collects a window of benign traffic, computes local reconstruction scores, and derives its detection threshold at a fixed quantile q . Training-phase and aggregation-phase defenses are structurally blind to this stage—they complete before calibration begins. If a single client’s benign calibration buffer can be contaminated with semantically benign but anomalously high (or low) reconstruction scores, the resulting threshold shifts without any change to model weights, training data, labels, or gradients. Such calibration-stage manipulation is structurally outside the scope of every defense that monitors the training or aggregation loop.

This paper. We identify threshold calibration as an under-analyzed attack surface and characterize its vulnerability under a score-level contamination model. Concretely, an adversary with access to a single client’s calibration buffer replaces a fraction f of entries with tail-sampled scores drawn from the same client’s benign distribution (*calibration-stage poisoning*). We evaluate three threshold-policy families—GLOBAL (federation-wide mean), LOCAL (per-client), and CLUSTER (cluster-averaged)—across all nine physical-device clients in the N-BaIoT dataset [15] at four injection fractions.

To preempt the natural objection—“this is just data poisoning applied to calibration data”—we argue that calibration-stage poisoning is structurally distinct along three axes. First, it is *temporally isolated*: it occurs strictly after model convergence, bypassing all training-phase defenses. Second, it exploits a *phase-specific trust model*: calibration data is collected autonomously by each client with no central validation gate, a weaker trust assumption than that applied to gradients or labels. Third, its effects are *policy-dependent* in ways that are not predictable from training-phase theory: the same injection fraction produces qualitatively different threshold shifts and spillover patterns depending on whether the policy aggregates thresholds globally, locally, or by cluster.

Contributions.

1. We isolate the post-training threshold-calibration stage as an undefended attack surface in federated IoT anomaly detection, distinct from training-phase and aggregation-phase poisoning (§3).
2. We present a controlled vulnerability characterization under score-level calibration contamination across three threshold policies, showing that the score-level raising attack reliably degrades victim detection rates by 2.4 to 7.9 percentage points across non-zero injection fractions (4.9–7.9 pp at $f=0.4$) with 95% bootstrap confidence intervals excluding zero (§5).
3. We document a policy-differentiated vulnerability profile: LOCAL and CLUSTER policies exhibit larger per-victim threshold shifts but no cross-client spillover; GLOBAL exhibits magnitude-diluted but federation-wide propagation; the score-level lowering attack succeeds for LOCAL and CLUSTER but is conditionally bounded for GLOBAL (§5).

Scope. All claims are bounded to the nine-device N-BaIoT setting with autoencoder detectors and FedAvg aggregation. The poisoning mechanism operates at the score level; end-to-end traffic-generation realizability is future work (§6). We do not claim deployability, evasion guarantees, or generalization beyond this setting. Defense design against calibration-stage poisoning is out of scope; we discuss properties a defense would need to satisfy in §6.

2 Background and Related Work

Federated anomaly detection for IoT. Network-level anomaly detection using federated learning allows devices to collaborate on a shared model while keeping raw traffic local [16,17]. Autoencoder-based reconstruction-error detectors are a standard approach: each client trains a local encoder-decoder on benign traffic; anomalies are flagged when reconstruction error exceeds a per-client detection threshold [21]. The N-BaIoT benchmark [15] provides labeled benign and Mirai/BASHLITE traffic for nine physical IoT device types, establishing a widely used evaluation platform for FL-IDS research.

Defenses. Existing defenses target the *training phase*. Byzantine-robust rules, including Krum [5], Trimmed-Mean, and Median [20], filter or downweight outlier gradients before aggregation. FLTrust [6] anchors aggregation to a trusted server dataset. Differential privacy approaches add noise to local updates [9]. These defenses all operate on gradients or model updates; they complete before the threshold-calibration stage begins.

Threshold calibration and poisoning. Threshold-based anomaly detection is sensitive to the calibration distribution: small shifts in the quantile-defining score set translate directly to missed detections or false alarms [12]. Poisoning attacks on anomaly detectors have been studied in centralized settings [11,7,2], but these assume either a single data owner or a trusted aggregator who can sanitize training data across the full dataset. In FL, calibration data never leaves the client, so server-side sanitization is not available. Prior work on FL model poisoning [1] and FL inference attacks [8] focuses on the gradient/weight surface. To our knowledge, no prior work characterizes the vulnerability of the post-training threshold-calibration stage in federated threshold-based anomaly detection.

Threshold policies. Federated threshold personalization spans a spectrum. GLOBAL policies share a single federation-mean threshold; LOCAL policies preserve per-client thresholds; CLUSTER policies group clients by calibration-score similarity and apply cluster-level thresholds [10,18]. The security implications of this design choice have not been studied under calibration-stage adversaries.

3 Threat Model

Attack surface. The threshold-calibration stage occurs after global model convergence and before threshold-dependent detection begins. Each client i collects a benign calibration set $\mathcal{C}_i = \{s_1, \dots, s_{n_i}\}$ of reconstruction scores from its own

traffic, then sets its local threshold $\tau_i = \text{quantile}(\mathcal{C}_i, q)$ at a fixed quantile q (here $q=0.95$). Depending on the federation’s threshold policy, τ_i may be used directly (LOCAL), averaged across clients (GLOBAL), or averaged within a calibration-score cluster (CLUSTER).

No existing FL training-phase or aggregation-phase defense monitors or validates the calibration buffer: it is collected autonomously by the client after the aggregation round completes.

Adversary model. The adversary controls a single compromised client v , which we call the *poisoned client*. Under LOCAL policy, only client v ’s own detection degrades; under GLOBAL policy, all federation members receive the contaminated threshold. We use “victim” to refer to client v itself when measuring per-client detection harm, and “non-victim clients” when referring to uninvolved federation members.

The adversary has *gray-box score-level access*: since the adversary controls the client, it can read and modify the local calibration buffer before threshold computation. The adversary does not access model weights, modify gradients, alter other clients’ data, or touch test data, test labels, or training labels. This is a strictly weaker capability than model-poisoning adversaries, which require gradient injection. The adversary operates at the score level; generating traffic that achieves a target reconstruction score is out of scope (end-to-end realizability is discussed in §6). The REPLACE-FIXED-BUDGET mechanism samples replacement scores with replacement from the tail reservoir; at high fractions ($f=0.4$), this may produce duplicate scores in the calibration buffer, which is detectable by a uniqueness check—this limitation is disclosed in §6.

Attack mechanics. We evaluate two attack objectives:

Raise. THRESHOLD-RAISE (primary): replace $m = \lfloor f \cdot n_v \rfloor$ randomly selected positions in $\mathcal{C}_v$ with scores drawn with replacement from the top-10% tail of $\mathcal{C}_v$ (HIGH-SCORE-BENIGN source). All replacement scores are semantically benign; AUROC is invariant by construction (test scores and labels are unchanged under poisoning). A raised threshold τ_v increases the detection threshold, suppressing alarms at the victim site.

Lower. THRESHOLD-LOWER (secondary, conditional): replace m positions with scores from the bottom-10% tail (LOW-SCORE-BENIGN). A lowered threshold increases the false-positive rate (alarm burden) and FPR disparity; the resulting $\Delta\text{TPR} > 0$ reflects threshold displacement and is a secondary indicator of operating-point shift. This attack is structurally conditional: random position replacement displaces upper-tail support with probability $\approx f$.

Negative control. RANDOM-BENIGN: replacement scores are sampled uniformly from the full calibration distribution rather than a tail. Any effect attributable to score contamination rather than position perturbation should vanish in this condition.

Scope boundary. This paper characterizes the vulnerability at the score level. A complete end-to-end attack would additionally require generating network traffic whose autoencoder reconstruction score lands in the target tail—this is future work (§6). The score-level characterization is a controlled exper-

iment, not a deployed attack. Claims are bounded to the single-client, single-compromise scenario; multi-client collusion is not evaluated.

4 Experimental Protocol

4.1 Dataset and Federation

We use N-BaIoT [15], a labeled IoT network traffic dataset covering nine physical device types (doorbells, baby monitor, security camera, smart TV, thermostat, webcam). Each device constitutes one federation client; all nine are eligible for calibration. Calibration set sizes range from 2,622 (Ecobee Thermostat) to 9,909 (Danmini Doorbell) benign samples per device (all ≥ 100 , satisfying the eligibility threshold). The federation trains a per-client autoencoder on benign traffic using FedAvg [14] for feature extraction, then enters the threshold-calibration stage. The full configuration is: threshold quantile $q=0.95$; injection fractions $f \in \{0.0, 0.1, 0.2, 0.4\}$; injection rule REPLACE-FIXED-BUDGET; single-client attack scope; 10 paired (training + poisoning) seeds; reservoir = bottom/top 10% tail of the poisoned client’s benign calibration scores, sampled with replacement.

4.2 Threshold Policies

Three threshold-policy families are evaluated, covering the main design points in the personalization literature:

Global. GLOBAL: $\tau_{\text{global}} = \frac{1}{N} \sum_{i=1}^N \tau_i^{\text{local}}$, where $\tau_i^{\text{local}} = \text{quantile}(\mathcal{C}_i, q)$. All clients receive the same threshold; any per-client shift is diluted by $1/N$.

Local. LOCAL: each client retains τ_i^{local} . Shifts are confined to the compromised client; no spillover.

Cluster. CLUSTER: clients are grouped by calibration-score similarity ($K=3$, $k\text{-means++}$, `random_state`= 42; assignments recomputed each seed as calibration fingerprints vary with training). Within each cluster, the shared threshold is the cluster mean; the higher seed-to-seed variance in CLUSTER CV(FPR) reflects cluster-assignment variability. A compromised client may alter the threshold for all co-cluster members (*intra-cluster spillover*); clients in other clusters are unaffected.

4.3 Attack Parameterization

For each (policy, attack objective, injection fraction $f \in \{0.0, 0.1, 0.2, 0.4\}$) combination, we run 10 paired seeds: each seed selects a training configuration and an independent poisoning draw. A clean copy of the calibration array is preserved; only the poisoned copy is used for threshold computation. The attack is applied to each of the nine clients as victim in turn (single-client scope), yielding 4 source-objective pairs $\times$ 3 threshold policies $\times$ 4 injection fractions $\times$ 9 victim clients $\times$ 10 seeds = 4,320 experimental cells total. Training seed $i \in \{0, \dots, 9\}$ is paired with poisoning seed $i+100$ and analysis seed $i+300$, giving statistically independent stochasticity across training, calibration sampling, and bootstrap resampling.

4.4 Statistical Analysis

Gate 1 (material shift, primary gate). Threshold shift $\Delta\tau_v = \tau_v^{\text{poisoned}} - \tau_v^{\text{clean}}$ is computed per seed. A seed is *Gate-1 significant* when $|\Delta\tau_v| > \delta\tau_v$ and the sign is consistent with the attack objective, where the materiality threshold is $\delta\tau_v = \max(0.1 \times \text{IQR}(\mathcal{C}_v^{\text{clean}}), 0.01 \times \text{median}_j(\text{IQR}(\mathcal{C}_j^{\text{clean}})))$. Gate-1 passes when $\geq 8/10$ seeds are significant and the victim-majority condition ($\geq 5/9$ victims significant) holds.

Gate 2 (directional excess over control). Bootstrap 95% CI on mean $\Delta\tau$ excludes zero for the primary strategy but spans zero for the negative control (RANDOM-BENIGN).

Gate 3 (downstream harm). Mean victim TPR change, ΔTPR , is directionally consistent (negative for raising, positive for lowering) across $\geq 8/10$ seeds.

Confidence intervals. All CIs are 95% percentile bootstrap over 10 seed-level aggregates, with $B=10,000$ resamples. AUROC invariance is verified as a protocol check: since test scores and labels are unchanged, AUROC must be identical under poisoning; any deviation would indicate a protocol violation.

4.5 Isolation Sanity Check

To confirm that the attack is confined to the calibration stage, we verify: (i) AUROC is invariant under poisoning for all cells, (ii) per-client rank ordering of clean anomaly scores is unchanged, and (iii) model weights before and after calibration are bitwise identical. All three hold in all 4320 cells.

5 Results

5.1 Clean-Baseline Policy Profile

Table 1 reports clean-baseline dispersion statistics across all ten seeds. LOCAL achieves the lowest CV(FPR) (0.333 ± 0.034), confirming that per-client thresholds suppress false-positive heterogeneity most effectively. GLOBAL exhibits the highest CV(FPR) (0.905 ± 0.039) because a single federation-wide threshold cannot simultaneously fit nine devices with different traffic distributions. CLUSTER lies in between (0.615 ± 0.143) with higher variance across seeds, reflecting sensitivity to cluster assignment. Bootstrap 95% CIs on the pairwise CV(FPR) difference exclude zero: Global–Local = 0.572 [0.537, 0.607]; Global–Cluster = 0.291 [0.214, 0.358]; Cluster–Local = 0.281 [0.201, 0.374], confirming that all three policies are statistically distinguishable on baseline fairness. Figure 2 shows the per-client FPR distributions for GLOBAL and LOCAL under the clean baseline.

5.2 Raising Attack: Primary Results

Table 2 and Figure 3(a) report threshold-shift and downstream detection harm for the HIGH-SCORE-BENIGN + THRESHOLD-RAISE strategy.

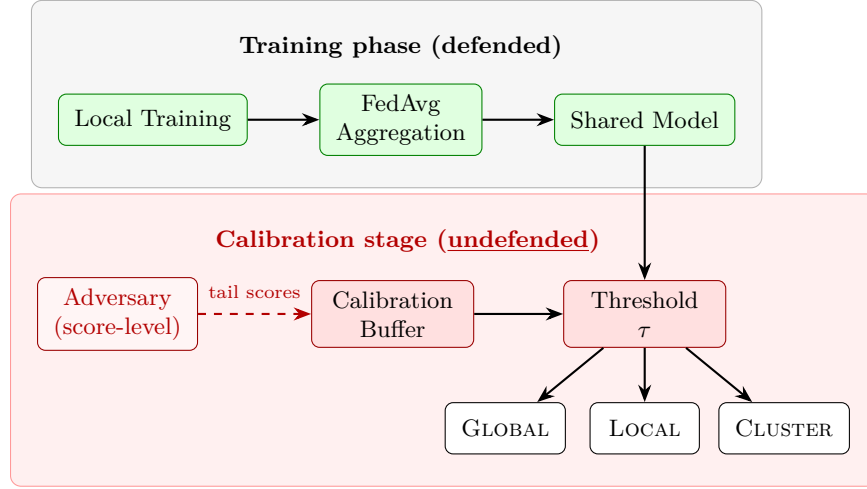


Fig. 1. Attack surface: the threshold-calibration stage occurs after FL training completes. Training-phase and aggregation-phase defenses (green) are structurally blind to the calibration stage (red). An adversary with score-level access to one client’s calibration buffer can shift τ without modifying model weights, gradients, or labels.

All three policies pass all three gates. The seed-level Gate-1 pass rate is 100% (10/10 seeds) at every non-zero fraction for all three policies. For LOCAL and CLUSTER, all nine clients are individually significant in every seed. For GLOBAL at $f=0.1$, two devices (Provision PT-737E and PT-838) are intermittently non-significant (2/10 and 4/10 seeds respectively), but the victim-majority condition ($\geq 5/9$ per seed) still holds because the remaining seven clients are consistently significant. Bootstrap CIs on $\Delta\tau$ exclude zero and remain non-overlapping with the RANDOM-BENIGN control (which produces CIs spanning zero at all fractions).

Policy-dependent magnitude. The raising attack reveals a clear policy-dependent magnitude profile. LOCAL and CLUSTER exhibit the largest threshold shifts: at $f=0.4$, $\Delta\tau = +0.902$ [$+0.864, +0.944$] and $+0.899$ [$+0.859, +0.941$] respectively. GLOBAL shows a structurally diluted but still significant shift: $+0.100$ [$+0.096, +0.105$] at $f=0.4$. The dilution is expected by construction—GLOBAL’s threshold is the federation mean, so a single-client shift is divided by $N=9$ —but the empirical question is whether the residual magnitude ($\approx 10\%$ of the local shift) suffices to produce measurable downstream harm. It does.

Downstream harm (Gate 3). Mean victim ΔTPR is negative and directionally consistent across all seeds for all three policies. At $f=0.4$: LOCAL -7.9 pp, CLUSTER -4.9 pp (plateauing from $f=0.2$), GLOBAL -6.1 pp. The GLOBAL harm (-6.1 pp) is notable: despite the 1/9-fold dilution of the per-client threshold shift, the federation-wide propagation of the poisoned threshold to *all* clients means that harm originating at the poisoned client propagates to all federation members. To contextualize: a victim with clean $\text{TPR} = 1.0$ (Dan-

mini_Doorbell) drops to ≈ 0.92 – 0.95 at $f=0.4$; a victim with clean TPR = 0.78 (Ennio_Doorbell) drops to ≈ 0.70 – 0.73 , translating to an additional ≈ 30 – 80 undetected Mirai flows per 1000 attack packets at representative traffic rates.

5.3 Lowering Attack: Secondary Results (Conditional)

Figure 3(b) shows threshold shift for the LOW-SCORE-BENIGN + THRESHOLD-LOWER strategy.

Local and Cluster succeed; Global is marginal. LOCAL achieves $\Delta\tau = -0.164$ $[-0.176, -0.154]$ at $f=0.4$ with Gate-3 $\Delta\text{TPR} = +11.8$ pp (detection-rate increase under the lowered threshold). CLUSTER similarly achieves -0.154 $[-0.167, -0.142]$ at $f=0.4$ with $\Delta\text{TPR} = +10.3$ pp. GLOBAL is structurally bounded: at $f=0.4$, $\Delta\tau = -0.018$ $[-0.020, -0.017]$, Gate-1 passes only at $f \geq 0.2$, and $\Delta\text{TPR} = +2.1$ pp.

FPR disparity (operational harm). The operational harm metric is $\Delta\text{CV}(\text{FPR})$ (change in coefficient of variation of per-client FPR, dimensionless): $+0.053$ (LOCAL, 10/10 seeds) and $+0.060$ (CLUSTER, 8/10 seeds) at $f=0.4$, while GLOBAL shows -0.019 (shared-threshold proportional FPR rise reduces relative disparity).

Mechanism and conditionality. Lowering a $q=0.95$ quantile requires displacing scores in the upper-tail support; random position replacement achieves this with probability $\approx f$ (40% at $f=0.4$), leaving 60% of upper-tail support intact. The GLOBAL policy compounds this via $1/N$ dilution. We present these results as a lower bound under REPLACE-FIXED-BUDGET; targeted upper-tail removal would achieve larger shifts.

5.4 Policy-Differentiation Summary

The empirical profile reveals three structurally distinct vulnerability signatures:

- **Local:** highest per-victim threshold shift magnitude; harm confined to the compromised client; no cross-client spillover; lowering attack effective.
- **Cluster:** near-LOCAL threshold shift magnitude; intra-cluster spillover observed (co-cluster clients share the contaminated threshold); lowering attack effective.
- **Global:** structurally diluted threshold shift ($\approx 1/N$ of local shift); but *full federation-wide spillover* (all 9 clients receive the poisoned threshold); raising attack produces significant victim harm despite dilution; lowering attack is marginal and condition-dependent.

A policy designer facing this trade-off must choose between isolated per-victim exposure (LOCAL) and diluted federation-wide propagation (GLOBAL), with CLUSTER offering intra-cluster but not inter-cluster spillover.

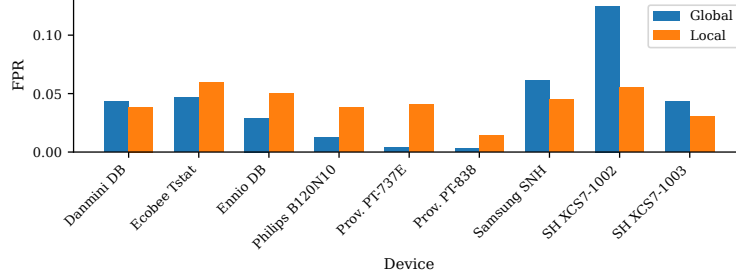


Fig. 2. Per-client FPR under clean baseline, GLOBAL vs. LOCAL threshold (seed 0). GLOBAL forces a single threshold on nine devices with heterogeneous score distributions, producing high CV(FPR) = 0.91 (seed 0). LOCAL fits each client’s own calibration distribution, achieving CV(FPR) = 0.33 on average.

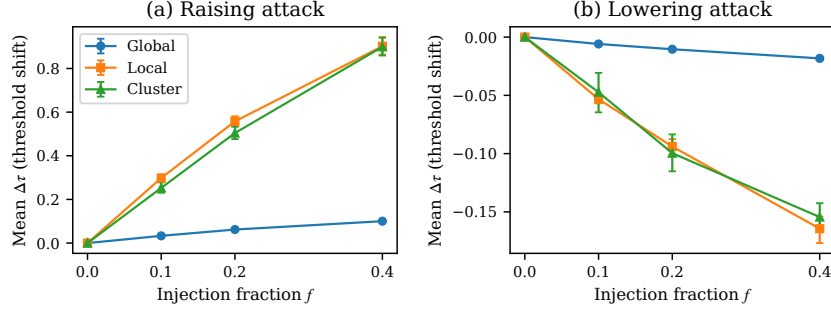


Fig. 3. Mean threshold shift $\Delta\tau$ vs. injection fraction f for the raising (a) and lowering (b) attacks across three policies. Error bars: 95 % bootstrap CI over 10 seeds. GLOBAL exhibits the smallest magnitude in both directions due to $1/N$ federation averaging. RANDOM-BENIGN negative control (not plotted) produces CI spanning zero at all fractions.

6 Discussion and Limitations

Score-level proxy and detectability. The experiment operates at the score level: reconstruction scores in the calibration buffer are replaced programmatically rather than through generated traffic. A complete end-to-end attack would additionally require generating network packets whose autoencoder reconstruction error lands in the target tail of the victim’s score distribution—a non-trivial requirement spanning feature engineering, the encoder-decoder nonlinearity, and physical network behavior. We present this work as a *vulnerability-surface characterization*, not a deployed attack; traffic-level realizability is future work. A related implementation limitation: with-replacement sampling from the tail reservoir at $f=0.4$ produces duplicate scores in the calibration buffer. With reservoir size $R \approx 10\% \cdot N$ and the smallest calibration set ($N \approx 2,622$), the expected colli-

Table 1. Clean-baseline dispersion statistics per threshold policy (N-BaIoT, $n=10$ seeds, 9 clients). CV: coefficient of variation (ddof=0); IQR: interquartile range. Lower CV(FPR) indicates more uniform false-positive rates across clients. Bold marks the best value per metric.

Policy	CV(FPR)	CV(TPR)	Worst BA	P10 Macro-F1
GLOBAL	0.905 ± 0.039	0.255 ± 0.030	0.657 ± 0.008	0.398 ± 0.053
LOCAL	0.333 ± 0.034	0.296 ± 0.036	0.655 ± 0.005	0.335 ± 0.058
CLUSTER	0.615 ± 0.143	0.313 ± 0.009	0.656 ± 0.004	0.299 ± 0.000

sion count $(N \cdot f)^2 / (2R) \approx 2,100$ expected collision pairs at $f=0.4$ —easily flagged by a uniqueness check; evading requires score perturbation at the cost of leaving the exact tail boundary. We flag detectability as an open problem.

GLOBAL spillover, lowering bounds, and policy implications. The $1/N$ -diluted shift is a definitional property of the GLOBAL policy; what is empirical is that the residual magnitude ($\approx 10\%$ of the local shift, $\Delta\text{TPR} = -6.1$ pp at $f=0.4$) remains measurable above the seed-to-seed variance. The lowering attack’s conditionality stems from the percentile mechanism: random position replacement displaces upper-tail support only with probability $\approx f$, leaving 60% of the upper tail intact at $f=0.4$. The Local/Cluster lowering results ($\Delta\text{CV}(\text{FPR}) = +0.05\text{--}0.06$ disparity increase, $\Delta\text{TPR} = +10\text{--}12$ pp detection-rate shift) are therefore a *lower bound* under the REPLACE-FIXED-BUDGET mechanism; adversaries with access to the buffer’s rank structure could achieve larger threshold displacements and correspondingly larger alarm-burden effects. These results together reveal a structural trade-off in policy design: LOCAL maximizes per-client isolation but maximizes per-victim exposure; GLOBAL limits per-victim shift via averaging but propagates residual harm federation-wide. Defense designers must choose which risk to bound.

Scope limitations. Claims are bounded to N-BaIoT (9 clients, autoencoder, FedAvg, $q=0.95$, calibration sets of ≥ 100 to ≈ 9900 benign samples per device). Generalizability to other anomaly detectors (isolation forest, LOF, VAE), FL algorithms (FedProx, SCAFFOLD), federation sizes, and IoT datasets is not established and is future work. The $n=10$ seed bootstrap CI coverage may differ from nominal at the tails. This paper does not propose or evaluate defenses. Any effective defense must address the calibration buffer as an unmonitored input channel, validate submitted thresholds against server-side priors, or impose calibration-set integrity constraints — requirements that differ from training-phase defense requirements and merit independent study.

Table 2. Threshold shift and detection harm under calibration-stage poisoning (N-BaIoT, $n=10$ seeds). $\Delta\tau$: mean threshold shift [95 % bootstrap CI]. G1: Gate-1 pass rate per seed (primary analysis: all 10/10 seeds). $\Delta\text{TPR}(\text{pp})$: victim detection-rate change; negative for Raise, positive for Lower (threshold displacement). Lowering rows ($\dagger$) are a lower bound; see §5 for $\Delta\text{CV}(\text{FPR})$ harm metric. RANDOM-BENIGN control: CI spans zero, $\text{G1} \leq 30\%$ for all conditions.

Policy	Attack	f	$\Delta\tau$ [95 % CI]	G1	$\Delta\text{TPR}(\text{pp})$
GLOBAL	Raise	0.1	+0.033 [+0.031, +0.035]	100 %	-2.4
		0.2	+0.062 [+0.059, +0.065]	100 %	-4.6
		0.4	+0.100 [+0.096, +0.105]	100 %	-6.1
LOCAL	Raise	0.1	+0.298 [+0.281, +0.314]	100 %	-6.0
		0.2	+0.558 [+0.535, +0.581]	100 %	-7.3
		0.4	+0.902 [+0.864, +0.944]	100 %	-7.9
CLUSTER	Raise	0.1	+0.253 [+0.231, +0.277]	100 %	-3.9
		0.2	+0.504 [+0.475, +0.532]	100 %	-4.9
		0.4	+0.899 [+0.859, +0.941]	100 %	-4.9
GLOBAL	Lower $\dagger$	0.1	-0.006 [-0.006, -0.006]	60 %	+0.8
		0.2	-0.010 [-0.011, -0.010]	100 %	+1.1
		0.4	-0.018 [-0.020, -0.017]	100 %	+2.1
LOCAL	Lower $\dagger$	0.1	-0.053 [-0.057, -0.049]	100 %	+2.8
		0.2	-0.094 [-0.100, -0.088]	100 %	+5.9
		0.4	-0.164 [-0.176, -0.154]	100 %	+11.8
CLUSTER	Lower $\dagger$	0.1	-0.047 [-0.065, -0.031]	100 %	+2.2
		0.2	-0.100 [-0.115, -0.084]	100 %	+5.4
		0.4	-0.154 [-0.167, -0.142]	100 %	+10.3

7 Conclusion

We characterized the post-training threshold-calibration stage in federated IoT anomaly detection as an attack surface that training-phase and aggregation-phase defenses structurally do not address. Under a score-level contamination model with a single compromised client, the raising attack degrades victim detection rates by 2.4–7.9 pp across non-zero injection fractions (4.9–7.9 pp at $f=0.4$) with 100 % Gate-1 pass rates and 95 % bootstrap CIs excluding zero. The lowering attack shifts detection thresholds downward, increasing victim TPR by 10–12 pp and $\Delta\text{CV}(\text{FPR})$ by 0.05–0.06 (as a lower bound under REPLACE-FIXED-BUDGET) for LOCAL and CLUSTER; both effects are operationally meaningful in IoT deployment contexts. The policy ordering—per-victim exposure maximized under LOCAL, diluted but federation-wide under GLOBAL, intra-cluster under CLUSTER—implies that calibration-stage hardening must match the aggregation design, not only the training-phase defense. Future work should address traffic-level realizability of the score-level injection model, calibration-stage integrity defenses, and generalization across anomaly-detector architectures and IoT datasets.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Bagdasaryan, E., Veit, A., Hua, Y., Estrin, D., Shmatikov, V.: How to backdoor federated learning. In: Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS). pp. 2938–2948. PMLR (2020)
2. Barreno, M., Nelson, B., Joseph, A.D., Tygar, J.D.: The security of machine learning. *Machine Learning* **81**(2), 121–148 (2010)
3. Bhagoji, A.N., Chakraborty, S., Mittal, P., Calo, S.: Analyzing federated learning through an adversarial lens. In: Proceedings of the 36th International Conference on Machine Learning (ICML). pp. 634–643. PMLR (2019)
4. Biggio, B., Nelson, B., Laskov, P.: Poisoning attacks against support vector machines. In: Proceedings of the 29th International Conference on Machine Learning (ICML). pp. 1467–1474 (2012)
5. Blanchard, P., El Mhamdi, E.M., Guerraoui, R., Stainer, J.: Machine learning with adversaries: Byzantine tolerant gradient descent. In: Advances in Neural Information Processing Systems (NeurIPS). vol. 30 (2017)
6. Cao, X., Fang, M., Liu, J., Gong, N.Z.: FLTrust: Byzantine-robust federated learning via trust bootstrapping. In: Network and Distributed System Security Symposium (NDSS) (2020)
7. Cretu, G.F., Stavrou, A., Locasto, M.E., Stolfo, S.J., Keromytis, A.D.: Casting out demons: Sanitizing training data for anomaly sensors. In: 2008 IEEE Symposium on Security and Privacy. pp. 81–95. IEEE (2008)
8. Geiping, J., Bauermeister, H., Dröge, H., Moeller, M.: Inverting gradients—how easy is it to break privacy in federated learning? In: Advances in Neural Information Processing Systems (NeurIPS). vol. 33, pp. 16937–16947 (2020)

9. Geyer, R.C., Klein, T., Nabi, M.: Differentially private federated learning: A client level perspective. In: *NeurIPS Workshop on Private Multi-Party Machine Learning* (2017)
10. Ghosh, A., Chung, J., Yin, D., Ramchandran, K.: An efficient framework for clustered federated learning. In: *Advances in Neural Information Processing Systems* (NeurIPS). vol. 33, pp. 19586–19597 (2020)
11. Kloft, M., Laskov, P.: Online anomaly detection under adversarial impact. In: *Proceedings of the 13th International Conference on Artificial Intelligence and Statistics (AISTATS)*. pp. 405–413. PMLR (2012)
12. L’heureux, A., Grolinger, K., Elyamany, H.F., Capretz, M.A.: Machine learning with big data: Challenges and approaches. *IEEE Access* **5**, 7776–7797 (2017)
13. Li, T., Sahu, A.K., Talwalkar, A., Smith, V.: Federated learning: Challenges, methods, and future directions. *IEEE Signal Processing Magazine* **37**(3), 50–60 (2020)
14. McMahan, B., Moore, E., Ramage, D., Hampson, S., y Arcas, B.A.: Communication-efficient learning of deep networks from decentralized data. In: *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*. pp. 1273–1282. PMLR (2017)
15. Meidan, Y., Bohadana, M., Mathov, Y., Mirsky, Y., Shabtai, A., Breitenbacher, D., Elovici, Y.: N-BaIoT: Network-based detection of IoT botnet attacks using deep autoencoders. *IEEE Pervasive Computing* **17**(3), 12–22 (2018)
16. Nguyen, D.C., Pham, Q.V., Pathirana, P.N., Ding, M., Seneviratne, A., Lin, Z., Dobre, O., Hwang, W.J.: Federated learning for smart healthcare: A survey. *ACM Computing Surveys* **55**(3), 1–37 (2022)
17. Rey, V., Sanchez Sanchez, P.M., Huertas Celdrán, A., Bovet, G.: Federated learning for malware detection in IoT devices. *Computer Networks* **204**, 108693 (2022)
18. Sattler, F., Müller, K.R., Samek, W.: Clustered federated learning: Model-agnostic distributed multitask optimization under privacy constraints. *IEEE Transactions on Neural Networks and Learning Systems* **32**(8), 3710–3722 (2020)
19. Tolpegin, V., Truex, S., Gursoy, M.E., Liu, L.: Data poisoning attacks against federated learning systems. In: *25th European Symposium on Research in Computer Security (ESORICS)*. pp. 480–501. Springer (2020)
20. Yin, D., Chen, Y., Kannan, R., Bartlett, P.: Byzantine-robust distributed learning: Towards optimal statistical rates. In: *Proceedings of the 35th International Conference on Machine Learning (ICML)*. pp. 5650–5659. PMLR (2018)
21. Zhou, C., Paffenroth, R.C.: Anomaly detection with robust deep autoencoders. In: *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. pp. 665–674 (2017)